SCOPE Retrofit Measure Analysis Report

Comprehensive Energy and Financial Analysis

Executive Summary

This report compares total site energy consumption, operational costs, construction costs, and payback period between baseline and all the applied renovation scenarios. The construction cost data is from 2026 RSMeans Database. Since RSMeans Database is proprietary, users can also adopt customized cost dataset when needed.

Building Information

- Weather File: USA_NY_Buffalo-Greater.Buffalo.Intl.AP.725280_TMY3.epw
- Building Name: SmallOffice
- Building Location: Buffalo, NY
- Total Floor Area: 511 m^2

Renovation Details by Scenario

Scenario	Renovation Type	Renovation Details
Baseline	baseline	**Current Condition of Exterior Wall:** • Exterior wall insulation: current R-value of insulation is R-11.3 **Current Condition of Roof:** • Roof insulation: current R-value of insulation is R-1.2 **Current Condition of Window:** • Number of window constructions is 20, with 20 simple glazing type windows, and 0 layered construction windows

		Current Condition of Door: • Number of door constructions is 3, with 1 glass door, 0 overhead door, and 2 door
Scenario 1	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Blown Mineral Wool, target R-20.4 **Roof upgrade:** • Roof insulation: Blown Mineral Wool, target R-34.5 **Window upgrade:** • 1-pane replacement • Weatherstrip application (skipped: no OperableWindow) • Glazing film application: safety film • Caulking application: acrylic • Window frame replacement: wood window frame **Door upgrade:** • Bottom seal: brush weatherstrip • Top/side seal: jamb weatherstrip
Scenario 2	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Blown Fiberglass, target R-20.4 **Roof upgrade:** • Roof insulation: Blown Fiberglass, target R-34.5 **Window upgrade:** • 2-pane replacement • Weatherstrip application (skipped: no OperableWindow) • Glazing film application: anti-graffiti film • Caulking application: polyurethane • Window frame replacement: wood window frame **Door upgrade:** • Bottom seal: silicone adhesive smoke gasket

		• Top/side seal: silicone adhesive smoke gasket
Scenario 3	wall + roof + window + door	**Wall upgrade:** • Exterior wall insulation: Fiberglass Batts, target R-20.4 **Roof upgrade:** • Roof insulation: Fiberglass Batts, target R-34.5 **Window upgrade:** • 3-pane replacement • Weatherstrip application (skipped: no OperableWindow) • Glazing film application: low-e film • Caulking application: polyurethane • Window frame replacement: wood-aluminium window frame **Door upgrade:** • Door replacement: polystyrene core steel door (1 of 3 door(s) skipped: door type incompatibility) • Bottom seal: automatic door bottom • Top/side seal: silicone adhesive smoke gasket

Annual Energy Analysis

Scenario	Metric	Baseline	Renovation	Delta	Savings %
Scenario 1	Anual Total Site Energy (GJ)	440.31	402.76	**37.55**	**8.53%**
Scenario 2	Anual Total Site Energy (GJ)	440.31	388.03	**52.28**	**11.87%**
Scenario 3	Anual Total Site Energy (GJ)	440.31	385.64	**54.67**	**12.42%**

Key Performance Metrics

Operational Cost Saving (Scenario - Baseline) ($/yr) Comparison

Baseline	$18,286
Scenario 3	$17,024

Baseline Scenario 3

Max saving: **$-1,262 (-6.90%)**

Total Site Energy Saving (Scenario - Baseline) (GJ/yr) Comparison

Baseline	440
Scenario 3	386

Baseline Scenario 3

Max saving: **55 GJ (12%)**

Min Retrofit Construction Cost

$152,898

Scenario 3

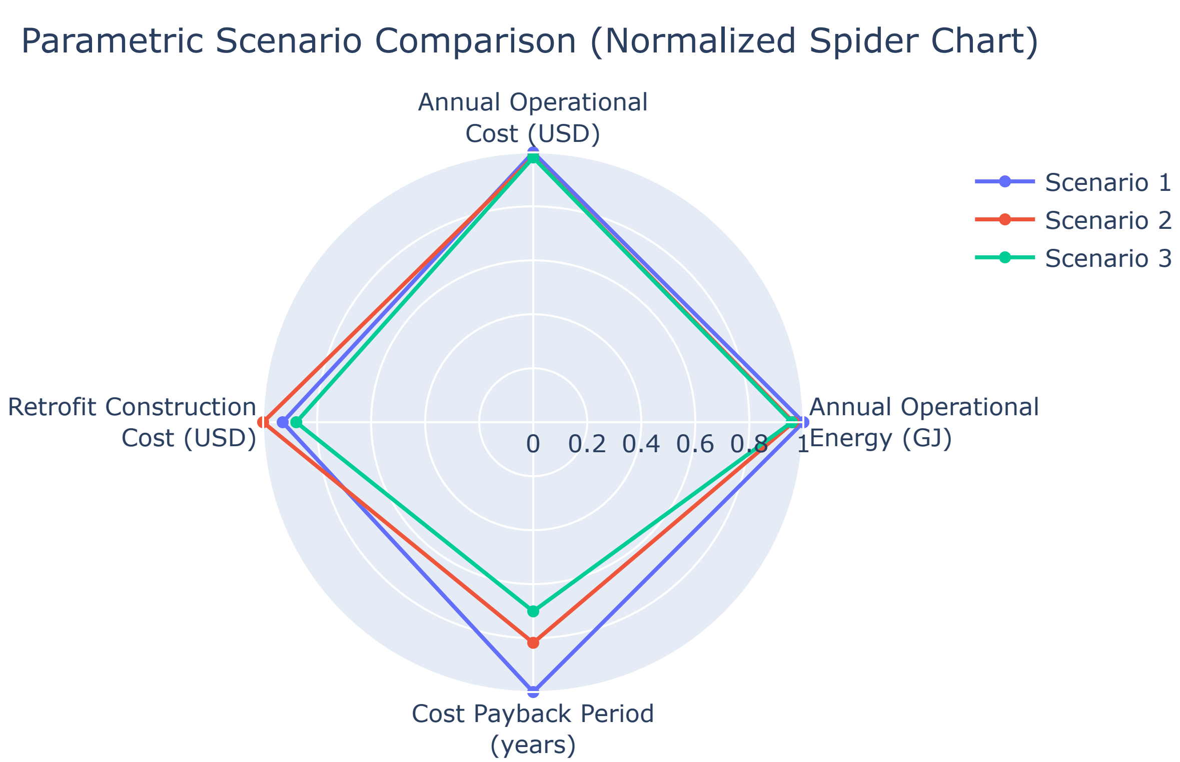
Lowest Retrofit Cost Payback Period

121 years

Scenario 3

Comparative Visualizations between Renovation Scenarios

Spider Chart Visualization



Payback Period of the Retrofit

Cost Payback Period

Payback = retrofit construction cost / abs(Operational Cost Saving (Scenario - Baseline) ($/yr)).

Scenario 1 173 yrs

Scenario 2 141 yrs

Scenario 3 121 yrs

Material List

> Material properties and consumption by renovation scenario. Missing values are shown as N/A.

Scenario	Component	Material name	Volume (m^3)	Area (m^2)	Thickness (m)	Length (m)	Thermal Conductivity (W/m·K)	Density (kg/m^3)	Product Lifetime (years)
Scenario 1	Wall insulation	Blown Mineral Wool	16	282	N/A	N/A	0.03	90	30
Scenario 1	Roof insulation	Blown Mineral Wool	144	599	0.24	N/A	0.03	90	30
Scenario 1	Door bottom sealing	brush weatherstrip	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 1	Door top/side sealing	jamb weatherstrip	N/A	N/A	N/A	16	N/A	N/A	15
Scenario 2	Wall insulation	Blown Fiberglass	17	282	N/A	N/A	0.03	30	30
Scenario 2	Roof insulation	Blown Fiberglass	144	599	0.24	N/A	0.03	30	30
Scenario 2	Door bottom sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 2	Door top/side sealing	silicone adhesive	N/A	N/A	N/A	16	N/A	N/A	15

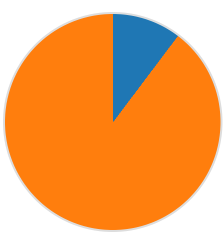
		smoke gasket							
Scenario 3	Wall insulation	Fiberglass Batts	17	282	N/A	N/A	0.03	30	30
Scenario 3	Roof insulation	Fiberglass Batts	144	599	0.24	N/A	0.03	30	30
Scenario 3	Door bottom sealing	automatic door bottom	N/A	N/A	N/A	3.66	N/A	N/A	15
Scenario 3	Door top/side sealing	silicone adhesive smoke gasket	N/A	N/A	N/A	16	N/A	N/A	15

Result Summary

Per-scenario breakdown: left chart shows retrofit construction cost contribution by measure, right chart shows total site energy by fuel type (electricity, natural gas, and other). Water is excluded from energy breakdown.

Scenario 1

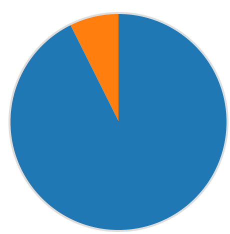
Cost Breakdown by Retrofit Measure



- Wall: $2,664 (10.3%)
- Roof: $23,226 (89.7%)

Total: $25,891

Annual Total Site Energy Breakdown (GJ)

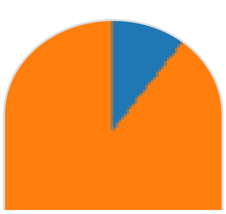


- Electricity: 373.07 GJ (92.6%)
- Natural Gas: 29.69 GJ (7.4%)

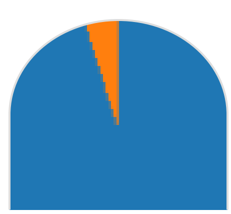
Total: 402.76 GJ

Scenario 2

Cost Breakdown by Retrofit Measure



Annual Total Site Energy Breakdown (GJ)



- Wall: $1,230 (10.6%)
- Roof: $10,401 (89.4%)

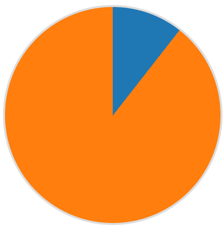
Total: $11,631

- Electricity: 369.41 GJ (95.2%)
- Natural Gas: 18.62 GJ (4.8%)

Total: 388.03 GJ

Scenario 3

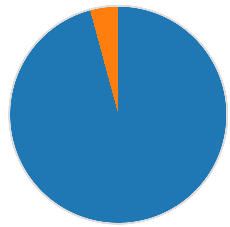
Cost Breakdown by Retrofit Measure



- Wall: $2,818 (10.6%)
- Roof: $23,820 (89.4%)

Total: $26,638

Annual Total Site Energy Breakdown (GJ)



- Electricity: 369.34 GJ (95.8%)
- Natural Gas: 16.30 GJ (4.2%)

Total: 385.64 GJ

Scenario	Retrofit Construction Cost (USD)	Annual Operational Cost (USD)	Anual Total Site Energy (GJ)
Baseline	$0.00	$18,286	440.31
Scenario 1	$161,670	$17,351	402.76
Scenario 2	$174,312	$17,052	388.03
Scenario 3	$152,898	$17,024	385.64

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